SpcTopics Assignment 1: Glucose Control Part 1

Addison Olstad

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1 Introduction

The toy model of glycemic control is a simplified mathematical representation of how the human body regulates blood glucose levels through a feedback variable that depends on both the current glucose deviation and its past values. We use this simplified model to investigate how the feedback parameters and external input affect the existence and stability of equilibria. The model is described by the following equations.

$$u = \alpha v + \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\, dt'$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t).$$

Here, v represents the deviation of glucose concentration from its baseline value, $\bar{v}$ represents the baseline glucose concentration, and u is a control variable that influences glucose regulation. The parameter γ represents the rate at which glucose is consumed by basic bodily functions, while $F(t)$ represents external influences on glucose concentration, such as food intake or physical exertion. The parameters α, β, λ, and δ determine the strength and dynamics of the control response.

The model incorporates a history-dependent control mechanism through the integral term in the first equation. This term calculates an exponentially weighted average of past glucose deviations, giving greater weight to more recent values. The parameter λ determines how quickly the influence of previous glucose deviations decays over time. The resulting control variable u affects glucose dynamics through the second equation, where the term $-\delta u(v + \bar{v})$ represents the control-dependent contribution to glucose removal.

2 Derive an ODE with 2 DOF

Starting with the toy control system, the integral must be differentiated for u, then replace the integral with the same equation.

Let:

$$I(t) = \int_{-\infty}^{t} \lambda e^{-\lambda(t-s)} v(s) ds$$

Then:

$$u = \alpha v + \beta I$$

Now, to differentiate the integral, the upper limit moves with t and the exponential inside also depends on t. Since $e^0 = 1$, and the remaining integral is $-\lambda I(t)$, let:

$$\dot{I} = \lambda v - \lambda I$$

Then differentiate u and substitute out I, simplify, and replace $\dot{v}$ with its given expression:

$$\dot{u} = -\lambda u + \lambda(\alpha + \beta)v + \alpha[-\gamma v - \delta u(v + \bar{v}) + F(t)]$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F$$

3 Reducing the Number of Parameters

Recall the reduced equations from the previous section, the number of parameters the equations will be nondimensionalized, using the scales below.

$$v = \bar{v}v' \to \frac{v}{\bar{v}} = v'$$

$$u = \tilde{u}u' \to \frac{u}{\tilde{u}} = u'$$

$$t = \tilde{t}t' \to \frac{t}{\tilde{t}} = t'$$

Scale the derivatives using the chain rule in the system of equations. First, with equation $\dot{v}$, plug the scales and simplify.

$$\frac{1}{\tilde{t}}\bar{v}\frac{dv'}{dt'} = -\gamma(\bar{v}v') - \delta(\tilde{u}u')(\bar{v}v' + \bar{v}) + F$$

$$\frac{dv'}{dt'} = -(\gamma\tilde{t})v' - (\delta\tilde{t}\tilde{u})u'(v' + 1) + \frac{\tilde{t}}{\bar{v}}F$$

Note: Let $G = \frac{\tilde{t}}{\bar{v}}F$
Rewrite $\gamma\tilde{t} = 1$ as $\tilde{t} = \frac{1}{\gamma}$, and $\delta\tilde{t}\tilde{u} = 1$ as $\tilde{u} = \frac{\gamma}{\delta}$.
Therefore:

$$\frac{dv'}{dt'} = -v' - u'(v' + 1) + G$$

As for the next equation, $\dot{u}$, with the chosen scales, the chain rule gives

$$\dot{u} = \frac{\tilde{u}}{\tilde{t}}\frac{du'}{dt'} = \frac{\gamma^2}{\delta}\frac{du'}{dt'}, \qquad \dot{v} = \frac{\bar{v}}{\tilde{t}}\frac{dv'}{dt'} = \gamma\bar{v}\frac{dv'}{dt'}.$$

Substituting these expressions gives

$$\frac{\gamma^2}{\delta}\frac{du'}{dt'} = -\frac{\lambda\gamma}{\delta}u' + \lambda(\alpha + \beta)\bar{v}v' + \alpha\gamma\bar{v}\frac{dv'}{dt'}.$$

Dividing every term by γ^2/δ yields

$$\frac{du'}{dt'} = -\frac{\lambda}{\gamma}u' + \frac{\delta\lambda\bar{v}}{\gamma^2}(\alpha + \beta)v' + \frac{\alpha\delta\bar{v}}{\gamma}\frac{dv'}{dt'}.$$

Let

$$a = \frac{\lambda}{\gamma}, \qquad b = \frac{\delta\lambda\bar{v}}{\gamma^2}(\alpha + \beta), \qquad c = \frac{\alpha\delta\bar{v}}{\gamma}.$$

Define the dimensionless glucose derivative by

$$f(u', v') = -v' - u'(v' + 1) + G.$$

Therefore, the dimensionless system is

$$\boxed{\begin{aligned}\frac{du'}{dt'} &= -au' + bv' + cf(u', v'),\\ \frac{dv'}{dt'} &= f(u', v').\end{aligned}}$$

4 Equilibrium Solutions

For the equilibrium analysis, assume that the external input $F(t) = F$ is constant. Consequently, the dimensionless input $G = F/(\gamma\bar{v})$ is also constant. Drop the primes for simplicity and set both time derivatives to zero.

$$-au + bv = 0$$
$$u = \frac{b}{a}v$$

$$-v - u(v+1) + G = 0$$
$$-v - \frac{b}{a}v(v+1) + G = 0$$

$$\boxed{v^* = -\frac{1}{2}\left(1+\frac{a}{b}\right) \pm \frac{1}{2}\sqrt{\left(1+\frac{a}{b}\right)^2 + \frac{4aG}{b}}, \qquad u^* = \frac{b}{a}v^*.}$$

The quadratic formula gives two branches. The real equilibria require the expression under the square root to be nonnegative. The branches meet when it is zero:

$$\boxed{G_{SN} = -\left(1+\frac{a}{b}\right)^2 \frac{b}{4a}.}$$

To determine local stability, consider the Jacobian of the dimensionless system, alongside the trace and determinant. For $\det J > 0$, a negative trace implies local asymptotic stability, while a positive trace implies instability. A negative determinant identifies a saddle.

For the simulation parameters $a = 0.1$ and $b = c = 1$, the branches meet at

$$G_{\mathrm{SN}} = -3.025, \qquad (u_{\mathrm{SN}}, v_{\mathrm{SN}}) = (-5.5, -0.55).$$

At this point,

$$J_{\mathrm{SN}} = \begin{pmatrix} -0.55 & 5.5 \\ -0.45 & 4.5 \end{pmatrix}, \qquad \det J_{\mathrm{SN}} = 0, \qquad \operatorname{tr} J_{\mathrm{SN}} = 3.95.$$

The eigenvalues are therefore 0 and 3.95: one eigenvalue vanishes at the fold, while the other equals the nonzero trace. The branches meet in a saddle-node bifurcation, with no real equilibria for $G < G_{\mathrm{SN}}$.

The existence of equilibria does not guarantee stability. For example, at $G = -3$, the upper equilibrium has two positive eigenvalues and is unstable. At $G = -1.5$, its eigenvalues have negative real parts, so it is locally asymptotically stable. At the base input $G = 1.6$, the upper equilibrium is approximately $(1.30074, 0.13007)$, with eigenvalues -0.44007 and -3.09074, whereas the lower equilibrium is a saddle.

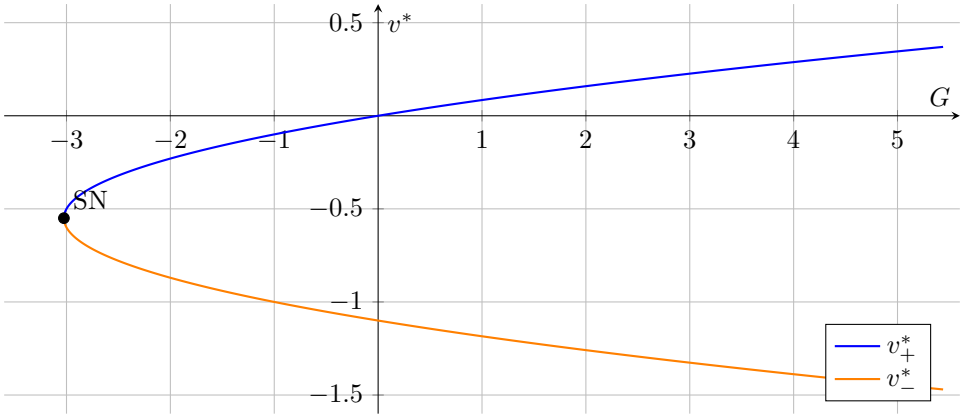


Figure 1: Equilibrium glucose-deviation branches for the simulation parameters $a = 0.1$ and $b = c = 1$. The branches meet at the saddle-node bifurcation $(G_{\mathrm{SN}}, v^*_{\mathrm{SN}}) = (-3.025, -0.55)$. There are no real equilibria for $G < -3.025$.

5 Exploring Initial Conditions

To investigate how the parameters and initial conditions affect recovery, a Python script was created to simulate trajectories of the dimensionless system. Here, recovery means convergence to a stable equilibrium, which doesn't need to coincide with the baseline glucose level. Numerically, an approach to an equilibrium is recorded when the distance from the equilibrium falls below 10^{-3}. Trajectories that cross the stopping threshold are recorded separately; crossing this threshold does not necessarily establish divergence.

Five numerical experiments were conducted to investigate how initial conditions and model parameters affect recovery toward a stable equilibrium:

1. Varied v_0 while fixing $u_0 = -8$ and all parameters, with finer sampling between $v_0 = -1.5$ and -1.0, where the observed outcomes changed.

2. Started trajectories near the upper equilibrium using small perturbations in u and v to examine whether they returned toward it.

3. Tested G below, at, and above $G_{\mathrm{SN}} = -3.025$ to distinguish equilibrium existence from recovery for a given initial condition.

4. Sampled G more closely between 3 and 5, keeping $(u_0, v_0) = (-8, -1.8)$, to locate changes in observed outcomes.

5. Varied a while fixing the remaining parameters and initial condition to examine its effect on recovery.

The five main experiments used second-order Runge–Kutta integration with step size $h = 0.001$ and maximum time $T = 40$. Proximity to a locally asymptotically stable equilibrium was checked every 100 steps. Runs stopped when the distance fell below 10^{-3}, when $v < -100$ or $\max(|u|, |v|) > 10^4$, or when the state became non-finite. Representative cases near the observed outcome changes were repeated with $h = 0.001$ and $h = 0.00025$, using a maximum time of $T = 200$.

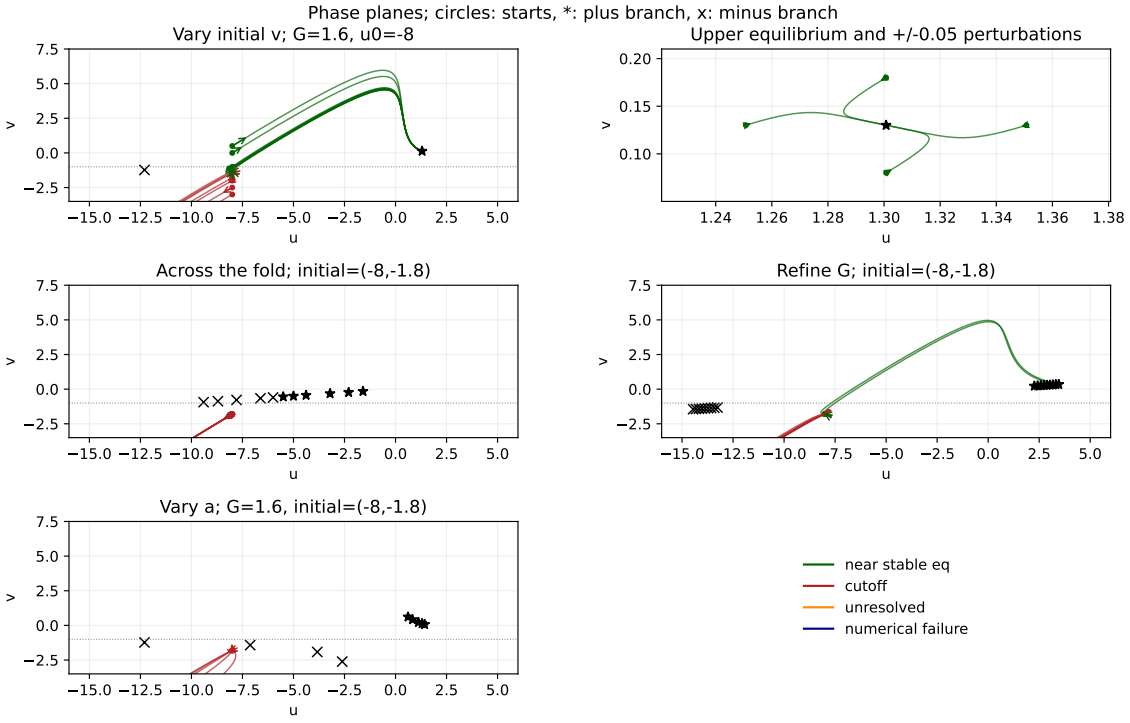


Figure 2: Phase-plane trajectories of the dimensionless glucose model. Unless varied, the parameters are $a = 0.1$, $b = c = 1$, and $G = 1.6$. (a) Varied v: $v_0 \in \{-3, -2.5, -2, -1.8, 0, 0.5\}$, together with $v_0 = -1.5, -1.45, \ldots, -1.0$. (b) Upper equilibrium, $(u^*, v^*) \approx (1.30074, 0.13007)$. (c) Input sweep across the fold $G_{\mathrm{SN}} = -3.025$, with $G \in \{-3.5, -3.1, -3.025, -3, -2.9, -2.5, -2, -1.5\}$. (d) Refined input sweep with $G = 3, 3.25, \ldots, 5$. (e) Decay-coefficient sweep with $a \in \{0.05, 0.1, 0.2, 0.5, 1\}$.

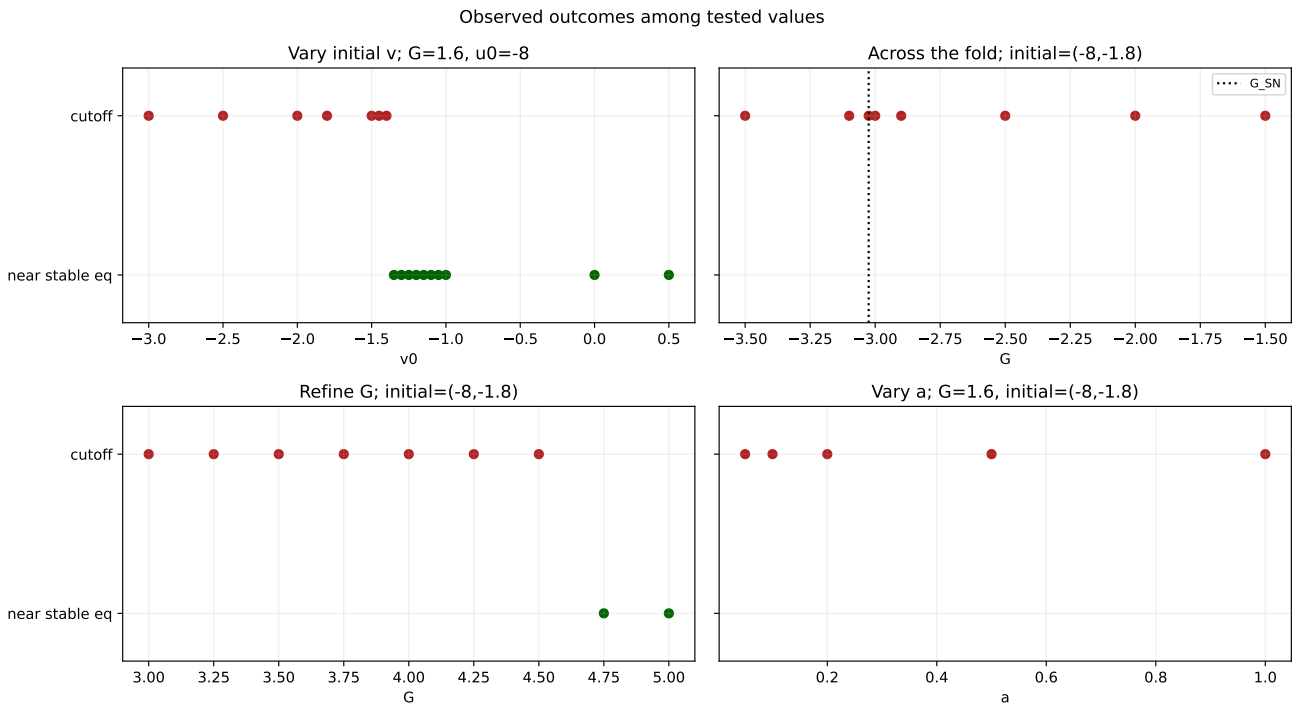


Figure 3: Observed trajectory outcomes among the tested initial conditions and parameter values, from said experiments listed above. Green points indicate that the trajectory reaches a distance below 10^{-3} from a locally asymptotically stable equilibrium; red points indicate crossing $v = -100$ or $\max(|u|, |v|) = 10^4$.

The phase portraits in Figure 2 show both recovery toward the upper equilibrium and departure toward the negative cutoff. Some recovering trajectories first go beyond the equilibrium glucose level before returning to it. The limits displayed demonstrate these transients and the initial departure of cutoff trajectories. At $a = 0.1$, $b = c = 1$, and $G = 1.6$, all trajectories starting from the tested small perturbations around the upper equilibrium returned to within 10^{-3} of it. This agrees with the negative eigenvalues obtained from the Jacobian. However, local stability does not imply recovery from every initial condition. With the same parameters, $(-8, -1)$ approached the upper equilibrium, but in contrast, $(-8, -1.5)$ reached the negative cutoff.

Figure 3 summarizes the observed dependence on initial conditions and parameters. In the refined initial-condition sweep, the outcome changed between $v_0 = -1.40$ and $v_0 = -1.35$. For $(u_0, v_0) = (-8, -1.8)$, the refined input sweep changed from a cutoff outcome at $G = 4.5$ to approach toward the stable equilibrium at $G = 4.75$. These results identify intervals containing changes in observed outcomes. Further sampling would be needed to locate these changes more precisely and determine whether each corresponds to a unique transition boundary.

All tested trajectories in the fold sweep reached the cutoff, including those with $G > G_{\mathrm{SN}}$. In particular, at $G = -1.5$, the upper equilibrium is locally asymptotically stable, yet the initial condition $(-8, -1.8)$ did not recover toward it. Thus, even the existence of a stable equilibrium does not guarantee recovery from a specified initial state. Likewise, all five tested decay coefficients, $a \in \{0.05, 0.1, 0.2, 0.5, 1\}$, produced cutoff outcomes for this initial condition at $G = 1.6$.

Overall, the simulations show that, even with the same parameters, some initial conditions lead to recovery while others reach the stopping cutoff. Changing the initial glucose deviation or the constant input can also change the outcome. Additionally, these experiments examine recovery with a constant input. They do not test whether a changing input could steer the system toward a desired state. The results apply only to the cases tested and do not identify all recovering initial conditions or exact boundaries between outcomes. Reaching the stopping cutoff does not prove that a trajectory diverges.